

PRAKHAR SHEKHAR PARTHASARTHI

LATENCY

-47%

P90 · AMAZON TRAVEL

Senior Software Engineer — Java · Distributed Systems · AWS

Three years at Amazon Travel.

Java microservices on AWS. Hotels, Flights, Insurance.
Ship services. Monitor p90. Keep on-call quiet.

CONV.

+33%

ON-CALL

20+ → 5

CPT

-43 bps

SERVICES

4 shipped

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SDE II / Amazon Travel — Hotels · Flights · Insurance

JUL 2023 – SEP 2025 · BENGALURU

Owned four Java microservices across the funnel. *Latency, resilience, quiet on-call.* Java 17 · Spring · AWS · CDK.

01 PARALLELIZED THE HOT PATH.

Refactored serial service fan-out to `CompletableFuture.allOf` with `bulkhead ExecutorService pools` per downstream. p90 fell **1.10s → 0.59s (-47%)**; conversion lifted **+33%**. No new hardware.

02 IDEMPOTENT CAPTURE ON SQS + STEP FUNCTIONS.

Dedupe via `DynamoDB idempotency keys` with TTL; dropped double-capture in renewals. CPT **-43 bps**; saved **~30 bps** of funnel-churn across insurance.

03 CDK-JAVA REWRITE OF DEPLOYS.

Replaced bespoke Python scripts with typed `CDK-Java` stacks and staged rollouts. Rollbacks **halved**, MTTR down **~1 day**, new services bootstrapped in hours.

04 DRAINED THE PAGER.

Rewrote alarms, killed config-hygiene SEVs **5-6 → 0**, on-call pages **20+ → ~5 / shift**. **150+ incidents** authored; stopped **~30 bps** of funnel-churn tied to silent failures.

02

SECTION TWO / RECORD

LEDGER · ARCHIVE · EVIDENCE

Prior roles, stack, projects, education, recognition. Terse. Load-bearing. Nothing decorative.

ROLE 02 / SDE I

SDE I — Amazon Travel

JUL 2022 — JUL 2023 · BENGALURU

Shipped REST services with **@Repository DynamoDB + Spring DI**. Owned the auto-insurance renewal path; **+60% renewal completion**. Held unit coverage at **80%+** across JUnit / JaCoCo. Quiet code, loud coverage.

ROLE 03 / INTERN

SDE Intern — Amazon Travel

MAY 2021 — FEB 2022 · BENGALURU

Led the **GraphQL → Java** migration on the renewals surface. Built **One-Click Renewal** end-to-end with **Guice DI**. Coverage held at **90%+**. Converted to full-time on the strength of it.

THE STACK

INSTRUMENTS · EIGHT CATEGORIES

JVM Java 17 · Spring Boot, MVC, DI · Guice · Maven · Gradle · strict nullability · pattern matching.	CONCURRENCY ExecutorService · CompletableFuture · ConcurrentHashMap · ScheduledExecutor · bulkhead pools.	AWS Lambda · DynamoDB · SQS · Step Functions · API Gateway · AppSync · CloudWatch · ECS · CDK-Java.	QUALITY JUnit 5 · Mockito · JaCoCo · integration harnesses · contract tests · golden-path fixtures.
FRONT-END React SSR & CSR · Redux · TypeScript · Vitest · accessible forms · edge-rendered bundles.	DATA PATTERNS DynamoDB single-table · idempotency keys with TTL · dedupe queues · outbox writes.	OPS Runbooks · alarm hygiene · SEV pages · on-call rotations · graceful degradation.	METHOD Trunk-based · code-review discipline · blameless postmortems · SLO-first design.

PROJECTS

OUT OF HOURS · EVIDENCE

Redis Server · in Java

github.com/praxstack/redis-server-java

Single-binary Redis-wire implementation — SET / GET / INCR / EXPIRE over RESP, event loop, AOF persistence. Benchmarks on commodity laptop; ~78% of C-Redis on my box.

SET 154K RPS	GET 195K RPS	INCR 128K	42 TESTS
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Open-Source Contributions

Open-source: 3 PRs merged into danielmiessler/Fabric (42k-star Go AI-tooling framework) — Bedrock bearer-token auth, dynamic region fetching, and a streaming-deadlock fix.

Markdown Viewer App

github.com/praxstack/markdown-viewer-app

Tiny Markdown reader. **Zero runtime dependencies**. Bundle <100 KB gzipped. Held by **155+ Vitest tests**; CI gate won't ship below threshold.

155+ TESTS	<100 KB GZ	ZERO DEPS
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EDUCATION · TRAINING

- **M.C.A. — Computer Science**
Motilal Nehru National Institute of Technology, Prayagraj
2019 — 2022 · GPA 9.25 / 10
- **B.Sc. (Hons) — Computer Science**
Bhaskaracharya College of Applied Sciences, Univ. of Delhi
PRAKHAR SHEKHAR PARTHASARTHI · CV · SWISS FLAG POSTER
2015 — 2018

RECOGNITION · CITATIONS

- **Amazon Pay — Merchants Categories Champion 2023**. Awarded for owning the renewals funnel across the merchant insurance line.
- **100+ trust-buster bugs** closed across travel quality sprints — contributed to **~30 bps of funnel-churn** stopped at source.